

MediSave Edge

**Linux-Based Medicine Storage Monitoring,
Inventory Alert and Redistribution Decision System**

Wipro COE Capstone Project

Linux | C++17 | Kernel Driver | IPC | Threads | TCP

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Introduction: Problem and Objectives

MediSave Edge is a Linux program that watches medicine storage, warns about expiry and low stock, and suggests how health facilities can share medicines.

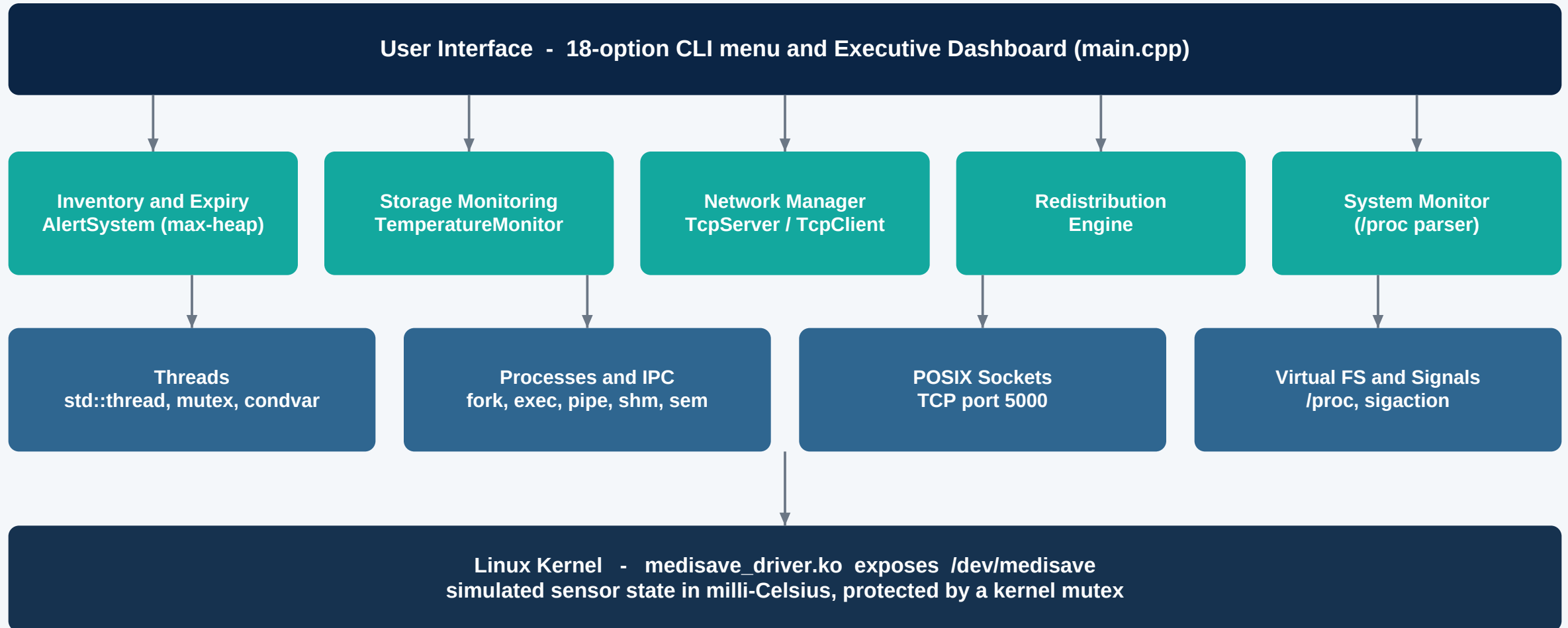
The Problem

- Medicines spoil if storage temperature goes wrong
- Medicines expire unused on the shelf
- One clinic has surplus while another has shortage
- Manual paper tracking gives no real-time alerts

What the Project Does

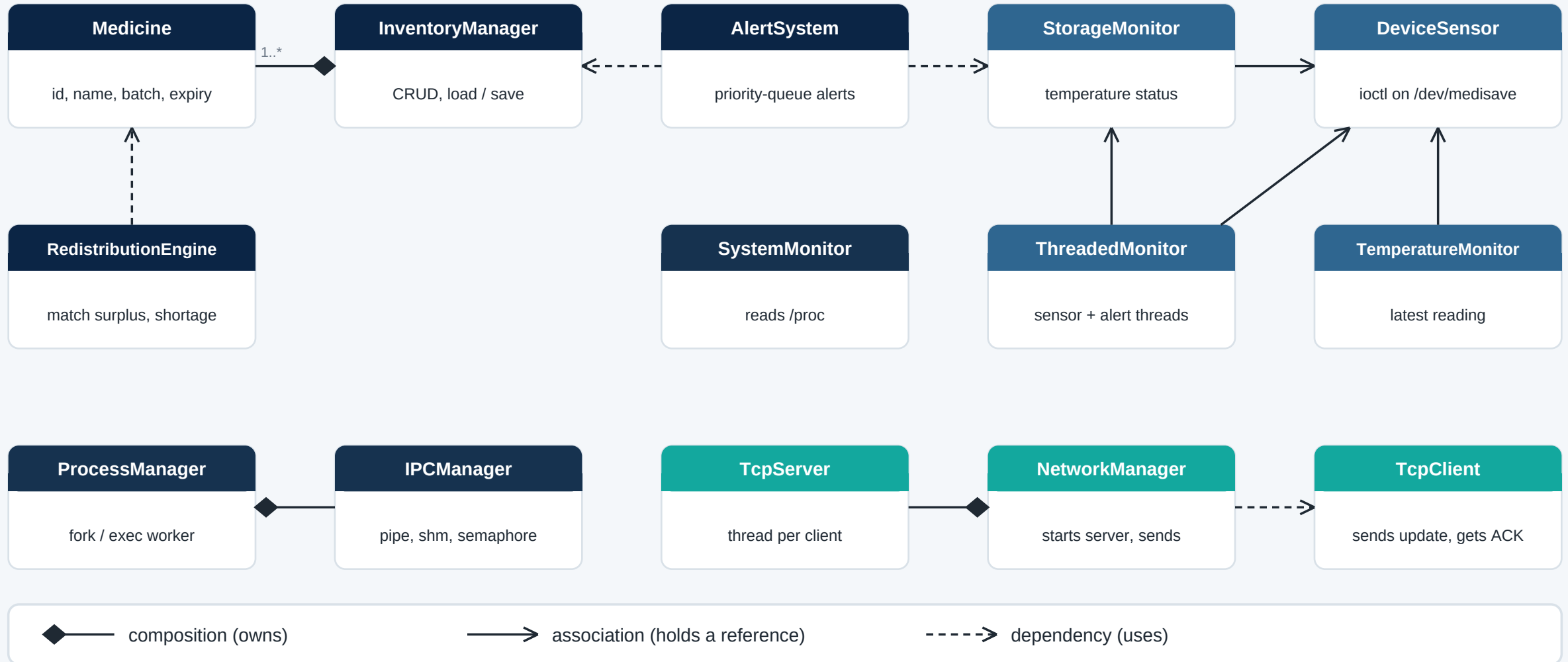
- Inventory management with file storage
- Expiry and low-stock alerts
- Temperature reading through a Linux kernel driver
- Background monitoring: processes, IPC, threads
- Facilities send updates over TCP
- Suggests medicine transfers; shows system health

System Architecture



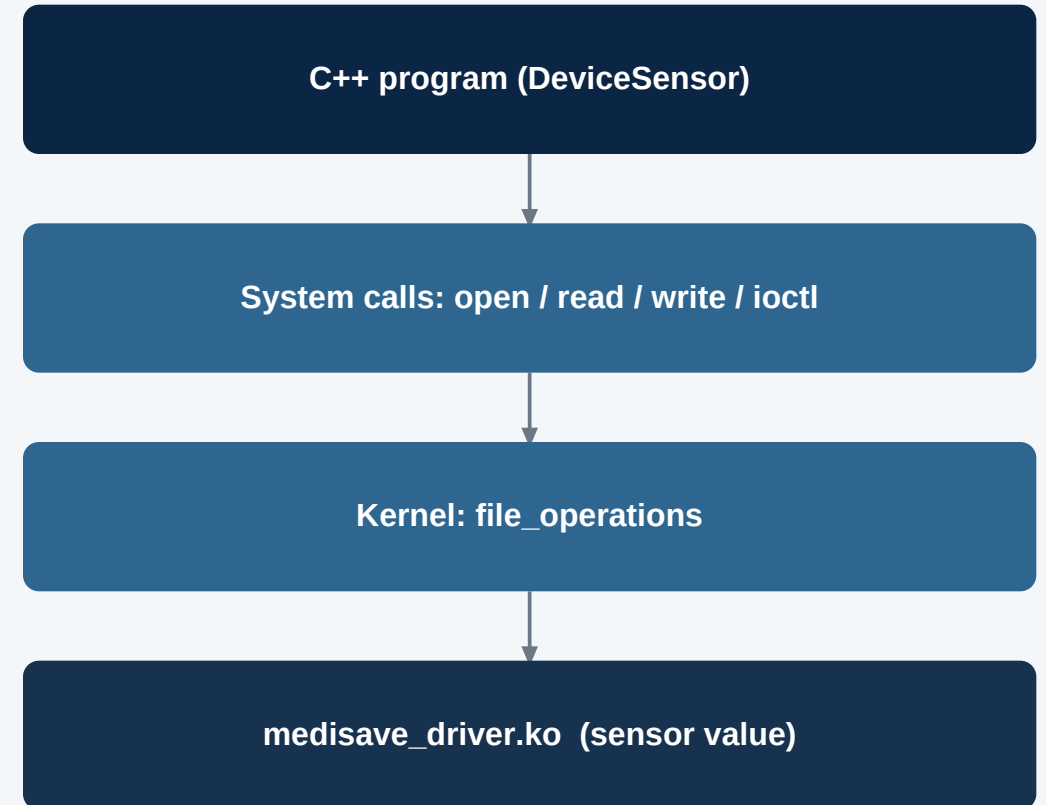
Each layer uses the layer below it: Interface > Modules > Linux features > Kernel driver.

UML Class Diagram



Linux Kernel Device Driver

- A kernel module that creates the device file `/dev/medisave`
- Programs talk to it using `open`, `read`, `write` and `ioctl`
- Temperature is stored as a whole number (milli-Celsius): 6500 means 6.50 C, because the kernel avoids floating point
- A kernel mutex protects the data from simultaneous access



LOW
below 2 C

NORMAL
2 - 8 C

WARNING
8 - 10 C

CRITICAL
above 10 C

Inventory Management

```
Terminal
$ ./bin/medisave (menu 5: Display Inventory)
```

ID	Medicine Name	Batch	Stock	Expiry Date	Status	Min/Max	Temp Range
MED-025	Cetirizine 10mg	CET-2028Z	450	2028-05-20	NORMAL	50/700	[15.0C to 25.0C]
MED-024	MMR Vaccine Live	MMR-2026M	12	2026-10-03	EXPIRED	20/120	[2.0C to 8.0C]
MED-023	Pantoprazole 40...	PAN-2027P	175	2027-08-30	NORMAL	30/350	[15.0C to 25.0C]
MED-022	BCG Vaccine Fre...	BCG-2026F	40	2027-02-14	NORMAL	15/100	[2.0C to 8.0C]
MED-021	Hydrocortisone ...	HYD-2026Y	14	2026-09-28	EXPIRED	20/150	[15.0C to 25.0C]
MED-019	Insulin Glargin...	GLA-2026G	30	2026-11-15	NORMAL	15/150	[2.0C to 8.0C]
MED-018	Amlodipine 5mg	AML-2027C	280	2028-03-22	NORMAL	50/500	[15.0C to 25.0C]
MED-017	Ciprofloxacin 5...	CIP-2026L	140	2026-10-28	WARNING	35/400	[15.0C to 25.0C]
MED-016	Tetanus Toxoid ...	TET-2026T	22	2026-10-20	WARNING	30/300	[2.0C to 8.0C]
MED-015	Omeprazole 20mg	OMP-2027W	320	2027-09-14	NORMAL	40/600	[15.0C to 25.0C]
MED-002	Amoxicillin 250mg	A2026B	12	2027-04-15	NORMAL	40/300	[15.0C to 25.0C]
MED-004	Ceftriaxone 1g Inj	CEF-2026K	60	2026-10-24	WARNING	25/250	[15.0C to 25.0C]
MED-003	Insulin Regular...	INS-2025X	45	2026-09-22	EXPIRED	20/200	[2.0C to 8.0C]
MED-009	Hepatitis B Vac...	HBV-2026V	18	2026-10-05	CRITICAL	25/150	[2.0C to 8.0C]
MED-014	Rabies Human Mo...	RAB-2026Q	5	2026-10-07	CRITICAL	10/50	[2.0C to 8.0C]
MED-010	Atorvastatin 20mg	ATV-2027H	150	2027-12-10	NORMAL	30/500	[15.0C to 25.0C]
MED-020	Doxycycline 100mg	DOX-2027X	95	2027-10-05	NORMAL	25/300	[15.0C to 25.0C]
MED-001	Paracetamol 500mg	P2026A	180	2026-10-04	CRITICAL	50/500	[15.0C to 25.0C]
MED-005	Ibuprofen 400mg	IBU-2026M	350	2027-11-30	NORMAL	50/600	[15.0C to 25.0C]
MED-006	Epinephrine Aut...	EPI-2026Z	8	2026-10-06	CRITICAL	15/100	[2.0C to 8.0C]
MED-011	Oxytocin 10IU Inj	OXY-2026J	10	2026-09-15	EXPIRED	15/200	[2.0C to 8.0C]
MED-008	Metformin 500mg	MET-2026E	400	2027-08-20	NORMAL	50/800	[15.0C to 25.0C]
MED-012	Salbutamol Inha...	SAL-2027S	85	2027-05-18	NORMAL	20/250	[15.0C to 25.0C]
MED-007	Azithromycin 500mg	AZI-2027D	220	2028-01-15	NORMAL	30/400	[15.0C to 25.0C]
MED-013	Oral Rehydratio...	ORS-2028R	500	2028-06-30	NORMAL	100/1000	[15.0C to 30.0C]

```
-----
Total distinct medicine records: 25 | Total cumulative stock units: 3621
```

Features

- Add, remove, update, search
- Update stock
- Data saved in data/medicines.txt

Status Colours

- NORMAL: safe
- WARNING: expires in 8-30 days
- CRITICAL: 0-7 days
- EXPIRED: past date

Expiry Report and Priority Alerts

```
Terminal
$ ./bin/medisave (menu 7: Check Expiry)
=====
          EXPIRY ANALYSIS REPORT
=====
Current System Reference Date: 2026-10-04

[EXPIRED MEDICINES] (Action Required: Quarantine / Dispose)
* Oxytocin 10IU Inj [ID: MED-011, Batch: OXY-2026J] Expired 19 days ago (2026-09-15) - Qty: 10
* MMR Vaccine Live [ID: MED-024, Batch: MMR-2026M] Expired 1 days ago (2026-10-03) - Qty: 12
* Hydrocortisone 100mg Inj [ID: MED-021, Batch: HYD-2026Y] Expired 6 days ago (2026-09-28) - Qty: 14
* Insulin Regular 100IU [ID: MED-003, Batch: INS-2025X] Expired 12 days ago (2026-09-22) - Qty: 45

[EXPIRING SOON - WITHIN 30 DAYS] (Action Required: Prioritize Consumption / Redistribution)
* [CRITICAL 0-7d] Epinephrine Auto-Inj [ID: MED-006, Batch: EPI-2026Z] Expires in 2 days (2026-10-06) - Qty: 8
* [CRITICAL 0-7d] Paracetamol 500mg [ID: MED-001, Batch: P2026A] Expires in 0 days (2026-10-04) - Qty: 180
* [CRITICAL 0-7d] Rabies Human Monoclonal [ID: MED-014, Batch: RAB-2026Q] Expires in 3 days (2026-10-07) - Qty: 5
* [WARNING 8-30d] Ceftriaxone 1g Inj [ID: MED-004, Batch: CEF-2026K] Expires in 20 days (2026-10-24) - Qty: 60
* [WARNING 8-30d] Ciprofloxacin 500mg [ID: MED-017, Batch: CIP-2026L] Expires in 24 days (2026-10-28) - Qty: 140
* [WARNING 8-30d] Tetanus Toxoid Vaccine [ID: MED-016, Batch: TET-2026T] Expires in 16 days (2026-10-20) - Qty: 22
* [CRITICAL 0-7d] Hepatitis B Vaccine [ID: MED-009, Batch: HBV-2026V] Expires in 1 days (2026-10-05) - Qty: 18
=====
```

- Expired batches are listed first
- CRITICAL = expires in 0-7 days, WARNING = 8-30 days

```
Terminal
$ ./bin/medisave (menu 8: Show Alerts)
=====
          MEDISAVE EDGE ALERTS
=====
[STORAGE NOTICE] Sensor driver not detected at /dev/medisave.
[CRITICAL] Oxytocin 10IU Inj
Batch: OXY-2026J
Status: EXPIRED 19 DAYS AGO
[CRITICAL] Insulin Regular 100IU
Batch: INS-2025X
Status: EXPIRED 12 DAYS AGO
[CRITICAL] Hydrocortisone 100mg Inj
Batch: HYD-2026Y
Status: EXPIRED 6 DAYS AGO
[CRITICAL] MMR Vaccine Live
Batch: MMR-2026M
Status: EXPIRED 1 DAYS AGO
[CRITICAL] Paracetamol 500mg
Batch: P2026A
Status: EXPIRES TODAY
```

Priority Queue

- Alerts are ranked
- Most urgent shown first

Processes, IPC and Threads

Processes

- `fork()` creates a worker, `execl()` runs `monitor_worker`
- `waitpid()` cleans it up (no zombie process)

IPC (between processes)

- Pipe sends messages from worker to main program
- Shared memory + semaphore share data safely

Threads (inside one program)

- One thread reads the sensor, one handles alerts
- mutex and condition variable keep data safe

```
Terminal
$ ./bin/process_test
[Monitor Worker] Started (PID: 118, Parent PID: 117, Polling Interval: 1s)
Parent PID : 117
Child PID  : 118
Child exited successfully.
[PASS] Process lifecycle
```

```
Terminal
$ ./bin/ipc_test
[PASS] Pipe creation
[PASS] Pipe communication
[PASS] Shared memory creation
[PASS] Shared memory communication
[PASS] Semaphore synchronization
[PASS] IPC cleanup
All IPC tests passed.
```

```
Terminal
$ ./bin/thread_test
[PASS] Sensor thread
[PASS] Mutex synchronization
[PASS] Condition variable
[PASS] Alert queue
[PASS] Thread shutdown
All thread tests passed.
```

TCP Networking and Redistribution

Terminal

```
$ ./bin/medisave_client Facility-A Paracetamol P2026A 150 SURPLUS
Sending: Facility-A|Paracetamol|P2026A|150|SURPLUS
Server acknowledgement received: ACK|Facility-A
Transaction completed successfully.
$ ./bin/medisave_client Facility-B Paracetamol P2026A 20 SHORTAGE
Sending: Facility-B|Paracetamol|P2026A|20|SHORTAGE
Server acknowledgement received: ACK|Facility-B
Transaction completed successfully.
```

How it works

- Facilities send updates to the server (port 5000)
- Message: FACILITY|MEDICINE|BATCH|QUANTITY|TYPE
- Server checks the message and replies ACK
- Engine matches a surplus with a shortage
- Transfer = smaller of surplus and shortage

Terminal

```
$ ./bin/medisave (menu 15: Analyze Redistribution)
=====
REDISTRIBUTION ANALYSIS
=====
Facilities received : 3
Stock records      : 27
Recommendations    : 1
-----
Recommendation #1
-----
Medicine : Paracetamol
Batch    : P2026A
Source Facility:
  Facility-A
  Available Surplus : 100 units
Destination Facility:
  Facility-B
  Required Shortage : 30 units
Suggested Transfer:
  30 units
Reason:
  Destination facility has a shortage while source facility has sufficient surplus.
-----
```

Linux System Monitoring

```
Terminal

$ ./bin/medisave (menu 16: Show System Health)
=====
                SYSTEM HEALTH
=====

CPU Model       : Intel(R) Xeon(R) Processor @ 2.10GHz
CPU Cores       : 1
CPU Usage       : 0.0 %

Memory Total    : 3.9 GB
Memory Used     : 0.2 GB
Memory Usage    : 5.1 %

System Uptime   : 0 minutes

Driver          : UNAVAILABLE
Storage Sensor  : UNAVAILABLE
```

How it works

- Reads Linux /proc files
- cpuinfo: processor model and cores
- stat: CPU usage
- meminfo: memory usage
- uptime: system running time

Driver shows UNAVAILABLE here because the kernel module was not loaded (insmod) in this run. The app still works without it.

Testing and Development Steps

```
Terminal

$ ./bin/test_inventory | tail -4
TEST RESULTS: 35 / 35 PASSED
$ ./bin/device_sensor_test | tail -4
TEST RESULTS: 14 / 14 PASSED
$ ./bin/ipc_test | tail -3
All IPC tests passed.
$ ./bin/process_test | tail -3
[PASS] Process lifecycle
$ ./bin/thread_test | tail -3
All thread tests passed.
$ ./bin/tcp_test | tail -3
All TCP tests passed.
$ ./bin/redistribution_test | tail -3
All redistribution tests passed.
$ ./bin/system_monitor_test | tail -3
All system monitor tests passed.
```

How it was built

- Day 1: Inventory and device driver
- Task 4: Driver integration with C++
- Task 5: Process and IPC
- Task 6: Multithreading and TCP
- Task 7: Redistribution, /proc monitor, dashboard
- All automated test suites pass

Thank You

Questions and discussion

MediSave Edge: Linux-Based Medicine Storage Monitoring and Redistribution